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-- Assignment III: Employee Management System - Table Creation with Data Types,
-- Constraints, and Indexes
-- Constraints:
-- Departments Table where:
-- * DepartmentID: Primary key
-- * DepartmentName: Unique and cannot be null
-- * Location: Optional text field
-- Employee table where
-- * EmployeeID: Primary key
-- * Email: Unique constraint
-- * Phone: Fixed 10-digit character
-- * Salary: Decimal with 2 precision and must be > 0
-- * HireDate: Default set to today's date
-- * Gender: Only 'M' or 'F' allowed
-- * FOREIGN KEY: Links to Departments
-- Create Indices
-- Index on LastName for faster search

-- CREATING DEPARTMENT SCHEMA
CREATE TABLE Department(
    dept_id INT PRIMARY KEY,
    dept_name VARCHAR(30) UNIQUE NOT NULL,
    location VARCHAR(50)
);

-- CREATING EMPLOYEE SCHEMA
CREATE TABLE Employee(
    emp_id INT PRIMARY KEY,
    emp_firstname VARCHAR(30),
    emp_lastname VARCHAR(30),
    email VARCHAR(50) UNIQUE,
    phn_no BIGINT CHECK(phn_no >= 1000000000 AND phn_no <= 9999999999),
    salary DECIMAL(10,2) CHECK(salary >= 0),
    hiring_date DATE DEFAULT (CURRENT_DATE),
    gender VARCHAR(10) CHECK(gender = 'M' OR gender = 'F'),
    dept_id INT,
    FOREIGN KEY (dept_id) REFERENCES Department(dept_id)
);

-- DESCRIBING TABLE TO CHECK THE OUTPUT
DESC Department;
DESC Employee;

-- INSERTING INTO DEPARTMENT TABLE
INSERT INTO Department(dept_id, dept_name, location) VALUES
    (1, 'CSE', 'Kolkata'),
    (2, 'CSE-AI/ML', 'Barasat'),
    (3, 'CSE-DS', 'Madhyamgram');

-- SHOWING ALL ROWS FROM DEPARTMENT
SELECT * FROM Department;

-- INSERTING INTO EMPLOYEE TABLE
INSERT INTO Employee(emp_id, emp_firstname, emp_lastname, email, phn_no, salary,
    hiring_date, gender, dept_id) VALUES
    (1, 'Sayantan', 'Bharati', 'sayantanbharati@gmail.com', 9876543210, 50000.25, CURDATE(), 'M', 1),
    (2, 'Anisha', 'Roy', 'anisharoy@gmail.com', 9844443210, 45000.25, CURDATE() + 1, 'F', 2),
    (3, 'Amitabh', 'Bacchan', 'amitabhbacchan@gmail.com', 9873333210, 87000.25, CURDATE() + 2, 'M', 2),
    (4, 'Einsine', 'Biswas', 'einsitebiswas@gmail.com', 9873999210, 74000.25, CURDATE(), 'M', 3),
    (5, 'Koyel', 'Bhattacharjee', 'koyel@gmail.com', 9873003210, 40000.25, CURDATE(), 'F', 3);

-- SHOWING ALL ROWS FROM EMPLOYEE
SELECT * FROM Employee;

-- CREATING INDEX ON LAST NAME
CREATE INDEX idx_lastname
ON Employee(emp_lastname);

-- SHOWING INDEX ON FOR EMPLOYEE TABLE
SHOW INDEX FROM Employee;

```

-- 1. Employee PRIMARY KEY violation

INSERT INTO Employee

(emp_id, emp_firstname, emp_lastname, email, phn_no, salary, hiring_date, gender, dept_id)

VALUES

(1, 'Rahul', 'Das', 'rahul@gmail.com', 9871111111, 30000.00, CURDATE(), 'M', 1);

-- 2. Department UNIQUE constraint violation

INSERT INTO Department

(dept_id, dept_name, location)

VALUES

(4, 'CSE', 'Howrah');

-- 3. Phone number CHECK constraint violation

-- Phone number has only 9 digits

INSERT INTO Employee

(emp_id, emp_firstname, emp_lastname, email, phn_no, salary, hiring_date, gender, dept_id)

VALUES

(6, 'Rahul', 'Das', 'rahul@gmail.com', 987654321, 30000.00, CURDATE(), 'M', 1);

-- 4. Salary less than 0

INSERT INTO Employee

(emp_id, emp_firstname, emp_lastname, email, phn_no, salary, hiring_date, gender, dept_id)

VALUES

(7, 'Ayan', 'Roy', 'ayan@gmail.com', 9871112222, -5000.00, CURDATE(), 'M', 1);

-- 5. Wrong gender input

INSERT INTO Employee

(emp_id, emp_firstname, emp_lastname, email, phn_no, salary, hiring_date, gender, dept_id)

VALUES

(9, 'Arjun', 'Pal', 'arjun@gmail.com', 9873334444, 40000.00, CURDATE(), 'X', 1);

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓	1 12:01:09	DROP DATABASE aug_3rd	2 row(s) affected	0.031 sec
✓	2 12:01:09	CREATE DATABASE aug_3rd	1 row(s) affected	0.000 sec
✓	3 12:01:09	USE aug_3rd	0 row(s) affected	0.000 sec
✓	4 12:01:09	CREATE TABLE Department(dept_id INT PRIMARY KEY, dept_name VARCHAR(30) UNIQUE NOT NULL, lo...	0 row(s) affected	0.031 sec
✓	5 12:01:09	CREATE TABLE Employee(emp_id INT PRIMARY KEY, emp_firstname VARCHAR(30), emp_lastname VARC...	0 row(s) affected	0.032 sec
✓	6 12:01:09	desc Department	3 row(s) returned	0.000 sec / 0.000 sec
✓	7 12:01:09	desc Employee	9 row(s) returned	0.000 sec / 0.000 sec
✓	8 12:01:09	INSERT INTO Department(dept_id,dept_name,location) VALUES (1,'CSE','Kolkata'), (2,'CSE-AI/ML','Barasa'...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
✓	9 12:01:09	SELECT * FROM Department LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
✓	10 12:01:09	INSERT INTO Employee(emp_id,emp_firstname,emp_lastname,email,phn_no,salary,hiring_date,gender,dept_id) VA...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
✓	11 12:01:09	SELECT * FROM Employee LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
✓	12 12:01:09	CREATE INDEX idx_lastname ON Employee(emp_lastname)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
✓	13 12:01:09	SHOW INDEX FROM Employee	4 row(s) returned	0.000 sec / 0.000 sec

Action Output

#	Time	Action	Message
✗	1 12:58:25	INSERT INTO Employee (emp_id, emp_firstname, emp_lastname, email, phn_no, salary, hiring_date, gender, dept_id) VALUES (1, 'Rahul', 'Das', 'rahul@gmail.com', '9871111...	Error Code: 1062. Duplicate entry '1' for key 'employee.PRIMARY'
✗	2 12:58:31	INSERT INTO Department (dept_id, dept_name, location) VALUES (4, 'CSE', 'Howrah')	Error Code: 1062. Duplicate entry 'CSE' for key 'department.dept_name'
✗	3 12:58:37	INSERT INTO Employee (emp_id, emp_firstname, emp_lastname, email, phn_no, salary, hiring_date, gender, dept_id) VALUES (6, 'Rahul', 'Das', 'rahul@gmail.com', '9876543...	Error Code: 3819. Check constraint 'employee_chk_1' is violated.
✗	4 12:58:44	INSERT INTO Employee (emp_id, emp_firstname, emp_lastname, email, phn_no, salary, hiring_date, gender, dept_id) VALUES (7, 'Ayan', 'Roy', 'ayan@gmail.com', '9871112...	Error Code: 3819. Check constraint 'employee_chk_2' is violated.
✗	5 12:58:50	INSERT INTO Employee (emp_id, emp_firstname, emp_lastname, email, phn_no, salary, hiring_date, gender, dept_id) VALUES (9, 'Arijun', 'Pal', 'arjun@gmail.com', '98733344...	Error Code: 3819. Check constraint 'employee_chk_3' is violated.

Result Grid

Filter Rows:

Edit

Export/Import:

Wrap Cell Content:

	emp_id	emp_firstname	emp_lastname	email	phn_no	salary	hiring_date	gender	dept_id
▶	1	Sayantan	Bharati	sayantanbharati@gmail.com	9876543210	50000.25	2026-08-09	M	1
	2	Anisha	Roy	anisharoy@gmail.com	9844443210	45000.25	2026-08-10	F	2
	3	Amitabh	Bacchan	amitabhbacchan@gmail.com	9873333210	87000.25	2026-08-11	M	2
	4	Einsine	Biswas	einsitebiswas@gmail.com	9873999210	74000.25	2026-08-09	M	3
	5	Koyel	Bhattacharjee	koyel@gmail.com	9873003210	40000.25	2026-08-09	F	3
✱	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Department Schema Employee Schema Inserting in Department Inserting in Employee Creating index in emp_lastname

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	Comment	Index_comment	Visible	Expression
▶	employee	0	PRIMARY	1	emp_id	A	3	NULL	NULL		BTREE			YES	NULL
	employee	0	email	1	email	A	3	NULL	NULL	YES	BTREE			YES	NULL
	employee	1	dept_id	1	dept_id	A	3	NULL	NULL	YES	BTREE			YES	NULL
	employee	1	idx_lastname	1	emp_lastname	A	5	NULL	NULL	YES	BTREE			YES	NULL

Department Schema Employee Schema Inserting in Department Inserting in Employee Creating index in emp_lastname

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Field	Type	Null	Key	Default	Extra
▶	dept_id	int	NO	PRI	NULL	
	dept_name	varchar(30)	NO	UNI	NULL	
	location	varchar(50)	YES		NULL	

Department Schema Employee Schema Inserting in Department Inserting in E

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Field	Type	Null	Key	Default	Extra
	emp_id	int	NO	PRI	NULL	
	emp_firstname	varchar(30)	YES		NULL	
	emp_lastname	varchar(30)	YES		NULL	
	email	varchar(50)	YES	UNI	NULL	
	phn_no	bigint	YES		NULL	
	salary	decimal(10,2)	YES		NULL	
	hiring_date	date	YES		NULL	
	gender	varchar(10)	YES		NULL	
▶	dept_id	int	YES	MUL	NULL	

Department Schema Employee Schema Inserting in Department Inserting in E

Result Grid

Filter Rows:

Edit

Export

	dept_id	dept_name	location
▶	1	CSE	Kolkata
	2	CSE-AI/ML	Barasat
	3	CSE-DS	Madhyamgram
✱	NULL	NULL	NULL

Department Schema Employee Schema Inserting in Department Inse